

Laboratory work #2

Normalization

Task 1

The table Booking_receipt violates 1NF because seat_numbers stores multiple values in one cell (e.g., "12A, 12B"), so the attribute is not atomic.

After fixing 1NF, it also violates 2NF: passenger data depends only on the passenger, not on the full composite key.

After fixing 2NF, it violates 3NF: departure_city depends on departure_airport_name, and arrival_city depends on arrival_airport_name — transitive dependencies, not direct dependencies on the key.

Task 2

Step 1: 1NF.

Split seat_numbers into separate rows. (one row per seat).

PK: booking_id, seat_number

Columns: passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price.

Step 2: 2NF (remove partial dependencies)

Passenger

- PK: passenger_id
- Columns: full_name, passport_number

Flight_info

- PK: flight_number
- Columns: departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name

Booking

- PK: booking_id
- FK: flight_number → Flight_info
- Columns: ticket_price

Booking_Passenger

- PK: booking_id, passenger_id
- FK: booking_id → Booking, passenger_id → Passenger
- Columns: seat_number

Step 3: 3NF (remove transitive dependencies)

Airport

- PK: airport_id
- Columns: airport_name, city

Airline

- PK: airline_id
- Columns: airline_name

Flight

- PK: flight_number
- FK: departure_airport_id → Airport, arrival_airport_id → Airport, airline_id →

Airline

Passenger

- PK: passenger_id
- Columns: full_name, passport_number

Booking

- PK: booking_id
- FK: flight_number → Flight
- Columns: ticket_price

Booking_Passenger

- PK: booking_id, passenger_id
- FK: booking_id → Booking, passenger_id → Passenger
- Columns: seat_number

All tables are now in 3NF — every non-key attribute depends only on the key, with no transitive dependencies.

Task 3

Functional dependencies

- booking_id → flight_number, ticket_price
- flight_number → departure_airport_id, arrival_airport_id, airline_id
- departure_airport_id → airport_name, city
- airline_id → airline_name
- passenger_id → full_name, passport_number
- (booking_id, passenger_id) → seat_number

Task 4

Checking flights table for 3NF

Flights(flight_id, sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at)

- Primary Key: flight_id

Functional Dependency: flight_id → all other attributes.

Analysis: departing_airport_id and airline_id are foreign keys; Flights does not store airport/airline names, so no transitive dependency exists inside the table. Gate values depend directly on flight_id, not on the airport.

Conclusion: the Flights table is in 3NF.

Task 5

```
CREATE TABLE Airport (  
    airport_id INT PRIMARY KEY,  
    airport_name VARCHAR(50) NOT NULL,  
    city VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Airline (  
    airline_id INT PRIMARY KEY,  
    airline_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Flight (  
    flight_number VARCHAR(10) PRIMARY KEY,  
    departure_airport_id INT NOT NULL REFERENCES Airport(airport_id),  
    arrival_airport_id INT NOT NULL REFERENCES Airport(airport_id),  
    airline_id INT NOT NULL REFERENCES Airline(airline_id)  
);
```

```
CREATE TABLE Passenger (  
    passenger_id INT PRIMARY KEY,  
    full_name VARCHAR(100) NOT NULL,  
    passport_number VARCHAR(20) NOT NULL  
);
```

```
CREATE TABLE Booking (  
    booking_id INT PRIMARY KEY,  
    flight_id INT NOT NULL REFERENCES Flight(flight_id),  
    passenger_id INT NOT NULL REFERENCES Passenger(passenger_id),  
    booking_time VARCHAR(50) NOT NULL  
);
```

```
    booking_id INT PRIMARY KEY,  
    flight_number VARCHAR(10) NOT NULL REFERENCES Flight(flight_number),  
    ticket_price DECIMAL(7,2) NOT NULL  
);
```

```
CREATE TABLE Booking_Passenger (  
    booking_id INT NOT NULL REFERENCES Booking(booking_id),  
    passenger_id INT NOT NULL REFERENCES Passenger(passenger_id),  
    seat_number VARCHAR(5) NOT NULL,  
    PRIMARY KEY (booking_id, passenger_id)  
);
```

-- INSERT

INSERT INTO Airport VALUES

```
(1, 'Almaty Intl', 'Almaty'),  
(2, 'Astana Intl', 'Astana'),  
(3, 'Istanbul Intl', 'Istanbul');
```

INSERT INTO Airline VALUES

```
(1, 'Air Astana'),  
(2, 'Turkish Airlines');
```

INSERT INTO Flight VALUES

```
('KC901', 1, 2, 1),  
( 'KC202', 2, 1, 1),  
( 'TK712', 1, 3, 2);
```

INSERT INTO Passenger VALUES

```
(1, 'Aigerim Zhaksybek', 'N1234567'),  
(2, 'Yerlan Nurlanov', 'N7654321'),  
(3, 'Dana Serik', 'N2233445'),  
(4, 'John Smith', 'P9988776'),  
(5, 'Aliya Bek', 'N5544332');
```

INSERT INTO Booking VALUES

```
(1001, 'KC901', 45000),  
(1002, 'KC202', 38000),  
(1003, 'TK712', 95000),
```

```
(1004, 'KC901', 45000);
```

```
INSERT INTO Booking_Passenger VALUES
```

```
(1001, 1, '12A'),
```

```
(1001, 2, '12B'),
```

```
(1002, 3, '5C'),
```

```
(1003, 4, '22A'),
```

```
(1004, 5, '13A');
```

Screenshots

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

Servers (1)

PostgreSQL 18

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airport_db

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phonebook_db

postgres

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Tablespaces

pgAgent Jobs

Query

Query History

airport_db/postgres@PostgreSQL 18

No limit

48 INSERT INTO Flight VALUES

49 ('KC901', 1, 2, 1),

50 ('KC202', 2, 1, 1),

51 ('TK712', 1, 3, 2);

52

53 INSERT INTO Passenger VALUES

54 (1, 'Aigerim Zhaksybek', 'N1234567'),

55 (2, 'Yerlan Nurlanov', 'N7654321'),

56 (3, 'Dana Serik', 'N2233445'),

57 (4, 'John Smith', 'P9988776'),

58 (5, 'Aliya Bek', 'N5544332');

59

60 INSERT INTO Booking VALUES

61 (1001, 'KC901', 45000),

62 (1002, 'KC202', 38000),

63 (1003, 'TK712', 95000),

64 (1004, 'KC901', 45000);

65

66 INSERT INTO Booking_Passenger VALUES

67 (1001, 1, '12A'),

68 (1001, 2, '12B'),

69 (1002, 3, '5C'),

70 (1003, 4, '22A'),

71 (1004, 5, '13A');

Data Output Messages Notifications

INSERT 0 5

Query returned successfully in 164 msec.

Total rows: Query complete 00:00:00.164

23°C Mostly sunny

11:19 23.09.2026

pgAdmin 4

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No limit

24

25 CREATE TABLE Booking (

26 booking_id INT PRIMARY KEY,

27 flight_number VARCHAR(10) NOT NULL REFERENCES Flight(flight_number),

28 ticket_price DECIMAL(7,2) NOT NULL

29);

30

31 CREATE TABLE Booking_Passenger (

32 booking_id INT NOT NULL REFERENCES Booking(booking_id),

33 passenger_id INT NOT NULL REFERENCES Passenger(passenger_id),

34 seat_number VARCHAR(5) NOT NULL,

35 PRIMARY KEY (booking_id, passenger_id)

36);

37

38

39

40 INSERT INTO Airport VALUES

41 (1, 'Almaty Intl', 'Almaty'),

42 (2, 'Astana Intl', 'Astana'),

43 (3, 'Istanbul Intl', 'Istanbul');

44

45 INSERT INTO Airline VALUES

46 (1, 'Air Astana'),

47 (2, 'Turkish Airlines');

Data Output Messages Notifications

INSERT 0 5

Query returned successfully in 164 msec.

Total rows: Query complete 00:00:00.164

CRLF Ln 38, Col 1

23°C Mostly sunny

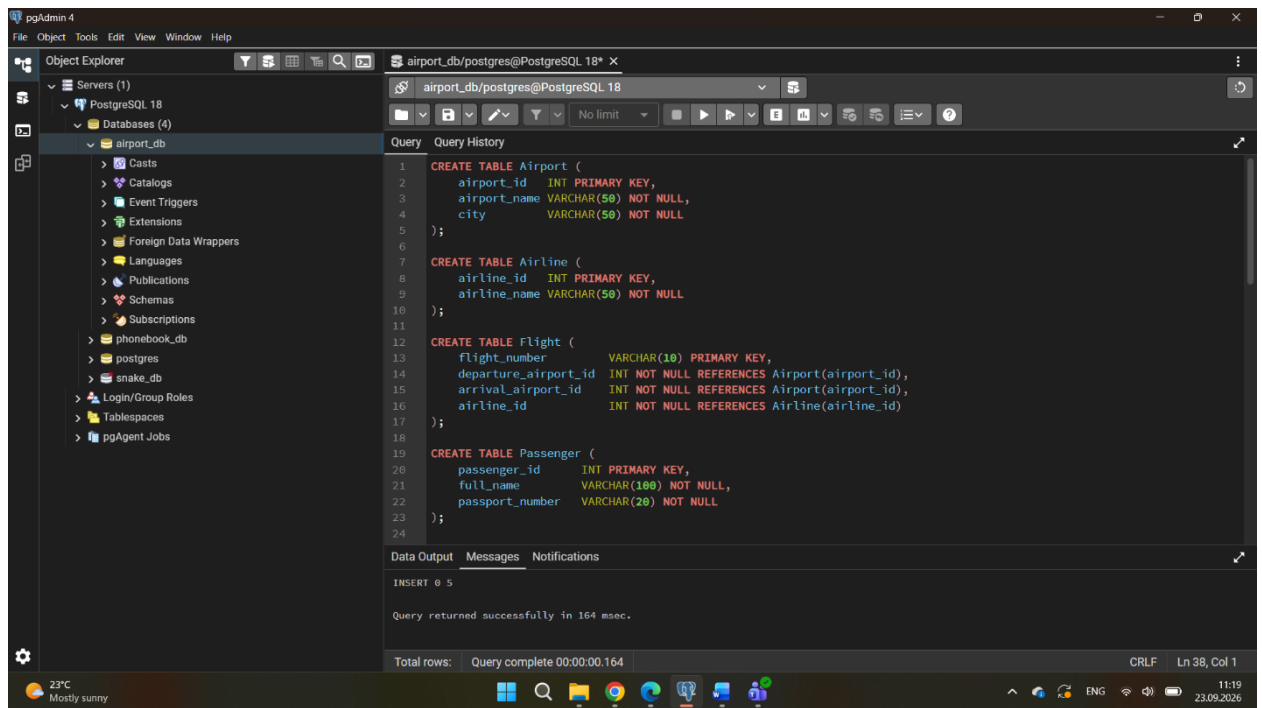
11:19 23.09.2026

Қайшы

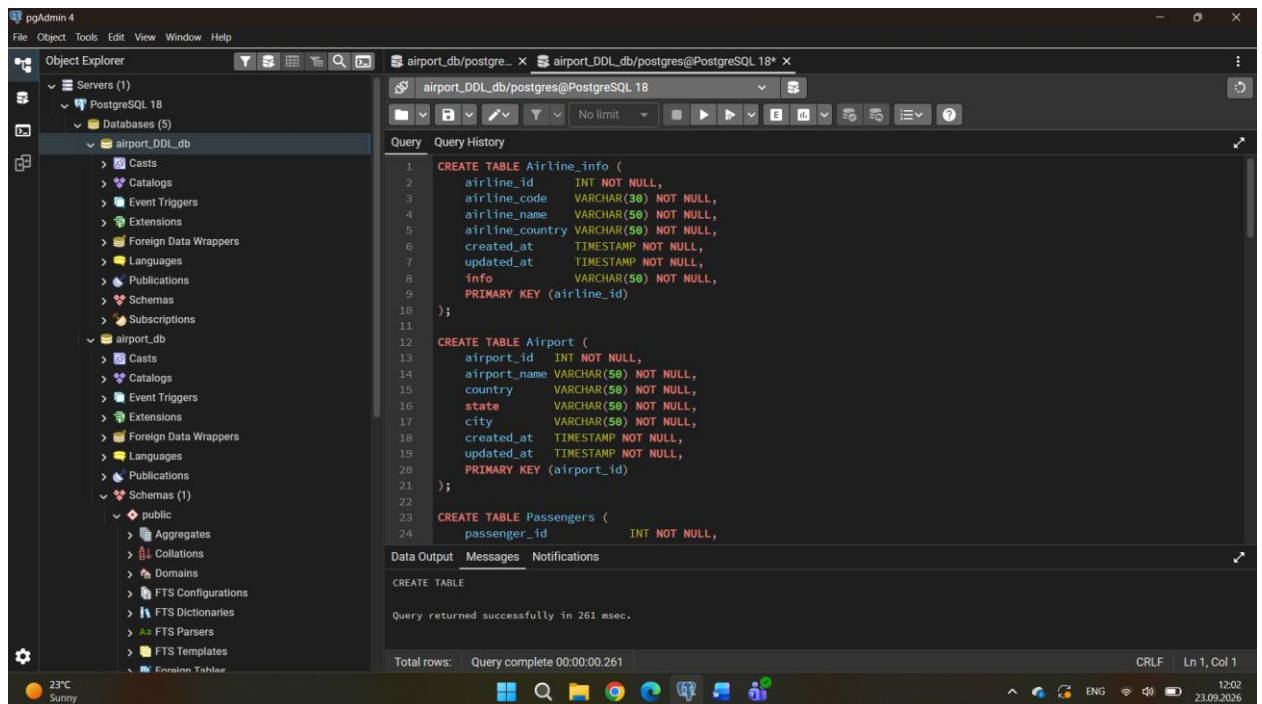
Скриншот аралық сақтағышқа көшірілді

"Скриншоттар" қалтасына автоматты түрде сақталды.

Белгілеу және бөлісу



DDL



Task 1: Create tables.

```

CREATE TABLE Airline_info (
    airline_id INT NOT NULL,
    airline_code VARCHAR(30) NOT NULL,
    airline_name VARCHAR(50) NOT NULL,

```

```
    airline_country VARCHAR(50) NOT NULL,  
    created_at      TIMESTAMP NOT NULL,  
    updated_at      TIMESTAMP NOT NULL,  
    info            VARCHAR(50) NOT NULL,  
    PRIMARY KEY (airline_id)  
);
```

```
CREATE TABLE Airport (  
    airport_id INT NOT NULL,  
    airport_name VARCHAR(50) NOT NULL,  
    country    VARCHAR(50) NOT NULL,  
    state      VARCHAR(50) NOT NULL,  
    city       VARCHAR(50) NOT NULL,  
    created_at TIMESTAMP NOT NULL,  
    updated_at TIMESTAMP NOT NULL,  
    PRIMARY KEY (airport_id)  
);
```

```
CREATE TABLE Passengers (  
    passenger_id      INT NOT NULL,  
    first_name        VARCHAR(50) NOT NULL,  
    last_name         VARCHAR(50) NOT NULL,  
    date_of_birth     DATE NOT NULL,  
    gender            VARCHAR(50) NOT NULL,  
    country_of_citizenship VARCHAR(50) NOT NULL,  
    country_of_residence VARCHAR(50) NOT NULL,  
    passport_number   VARCHAR(20) NOT NULL,  
    created_at        TIMESTAMP NOT NULL,  
    updated_at        TIMESTAMP NOT NULL,  
    PRIMARY KEY (passenger_id)  
);
```

```
CREATE TABLE Flights (  
    flight_id      INT NOT NULL,  
    sch_departure_time TIMESTAMP NOT NULL,  
    sch_arrival_time  TIMESTAMP NOT NULL,  
    departing_airport_id INT NOT NULL,  
    arriving_airport_id INT NOT NULL,
```

```
departing_gate    VARCHAR(50) NOT NULL,  
arriving_gate     VARCHAR(50) NOT NULL,  
airline_id        INT NOT NULL,  
act_departure_time  TIMESTAMP NOT NULL,  
act_arrival_time   TIMESTAMP NOT NULL,  
created_at         TIMESTAMP NOT NULL,  
updated_at         TIMESTAMP NOT NULL,  
PRIMARY KEY (flight_id)  
);
```

```
CREATE TABLE Booking (  
    booking_id     INT NOT NULL,  
    flight_id      INT NOT NULL,  
    passenger_id   INT NOT NULL,  
    booking_platform VARCHAR(50) NOT NULL,  
    created_at     TIMESTAMP NOT NULL,  
    updated_at     TIMESTAMP NOT NULL,  
    status         VARCHAR(50) NOT NULL,  
    price          DECIMAL(7,2) NOT NULL,  
    PRIMARY KEY (booking_id)  
);
```

```
CREATE TABLE Booking_flight (  
    booking_flight_id INT NOT NULL,  
    booking_id        INT NOT NULL,  
    flight_id         INT NOT NULL,  
    created_at        TIMESTAMP NOT NULL,  
    updated_at        TIMESTAMP NOT NULL,  
    PRIMARY KEY (booking_flight_id)  
);
```

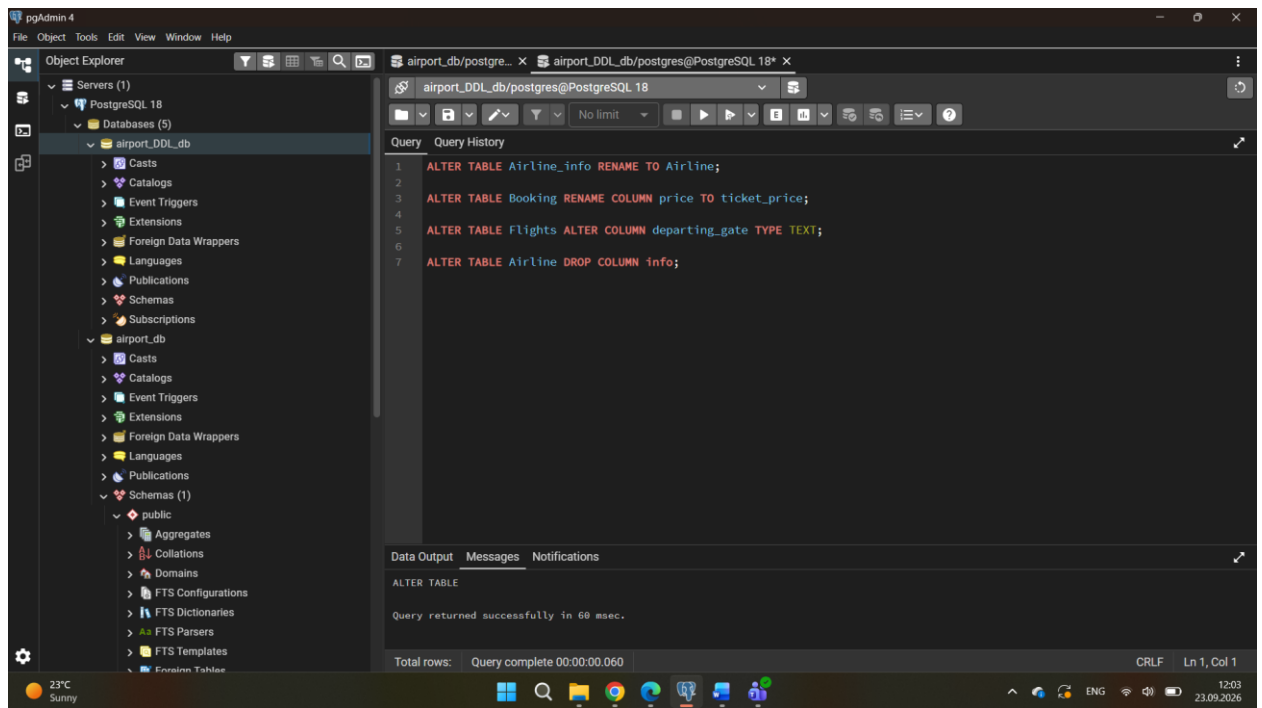
```
CREATE TABLE Boarding_pass (  
    boarding_pass_id INT NOT NULL,  
    booking_id       INT NOT NULL,  
    seat             VARCHAR(50) NOT NULL,  
    boarding_time    TIMESTAMP NOT NULL,  
    created_at       TIMESTAMP NOT NULL,  
    updated_at       TIMESTAMP NOT NULL,
```

```
PRIMARY KEY (boarding_pass_id)
);
```

```
CREATE TABLE Baggage (
  baggage_id INT NOT NULL,
  weight_in_kg DECIMAL(4,2) NOT NULL,
  created_at TIMESTAMP NOT NULL,
  updated_at TIMESTAMP NOT NULL,
  booking_id INT NOT NULL,
  PRIMARY KEY (baggage_id)
);
```

```
CREATE TABLE Baggage_check (
  baggage_check_id INT NOT NULL,
  check_result VARCHAR(50) NOT NULL,
  created_at TIMESTAMP NOT NULL,
  updated_at TIMESTAMP NOT NULL,
  booking_id INT NOT NULL,
  passenger_id INT NOT NULL,
  PRIMARY KEY (baggage_check_id)
);
```

```
CREATE TABLE Security_check (
  security_check_id INT NOT NULL,
  check_result VARCHAR(20) NOT NULL,
  created_at TIMESTAMP NOT NULL,
  updated_at TIMESTAMP NOT NULL,
  passenger_id INT NOT NULL,
  PRIMARY KEY (security_check_id)
);
```



4-7.

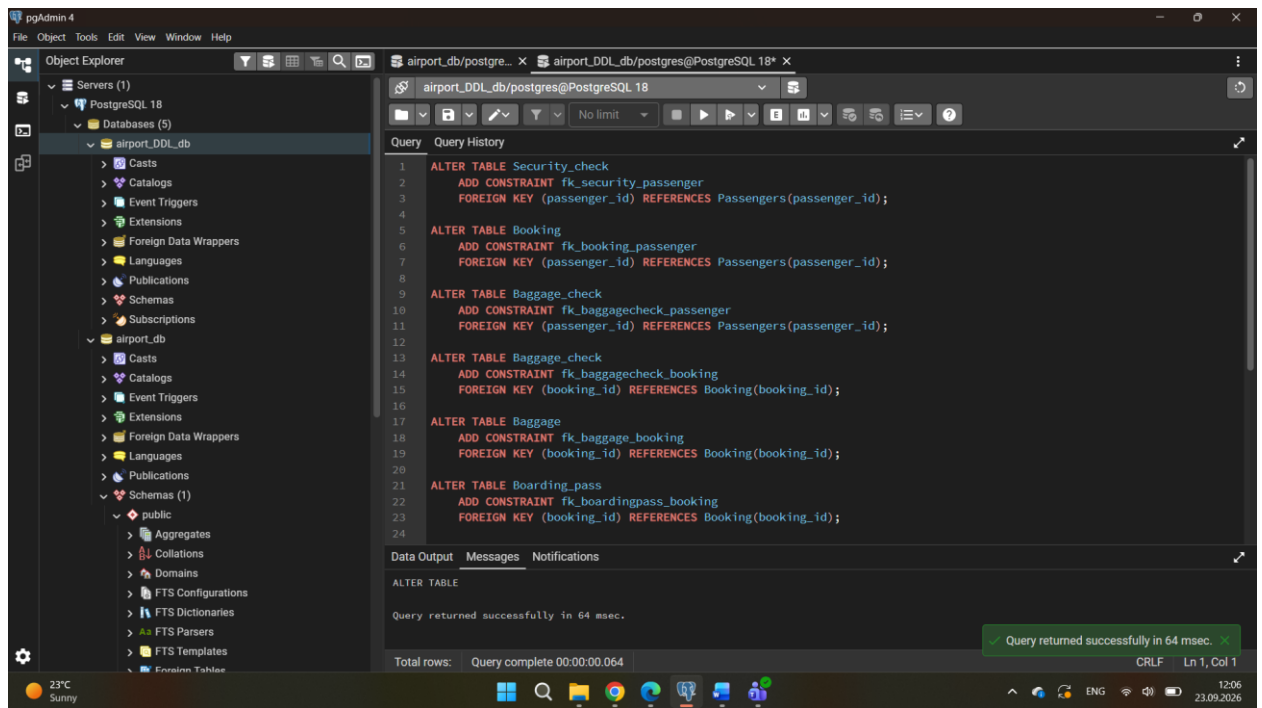
ALTER TABLE Airline_info RENAME TO Airline;

ALTER TABLE Booking RENAME COLUMN price TO ticket_price;

ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;

ALTER TABLE Airline DROP COLUMN info;

;



Foreign keys

ALTER TABLE Security_check

ADD CONSTRAINT fk_security_passenger

FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Booking

ADD CONSTRAINT fk_booking_passenger

FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Baggage_check

ADD CONSTRAINT fk_baggagecheck_passenger

FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);

ALTER TABLE Baggage_check

ADD CONSTRAINT fk_baggagecheck_booking

FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

ALTER TABLE Baggage

ADD CONSTRAINT fk_baggage_booking

FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);

ALTER TABLE Boarding_pass

ADD CONSTRAINT fk_boardingpass_booking

```
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
```

```
ALTER TABLE Booking_flight  
  ADD CONSTRAINT fk_bookingflight_booking  
  FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
```

```
ALTER TABLE Booking_flight  
  ADD CONSTRAINT fk_bookingflight_flight  
  FOREIGN KEY (flight_id) REFERENCES Flights(flight_id);
```

```
ALTER TABLE Flights  
  ADD CONSTRAINT fk_flights_departing_airport  
  FOREIGN KEY (departing_airport_id) REFERENCES Airport(airport_id);
```

```
ALTER TABLE Flights  
  ADD CONSTRAINT fk_flights_arriving_airport  
  FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id);
```

```
ALTER TABLE Flights  
  ADD CONSTRAINT fk_flights_airline  
  FOREIGN KEY (airline_id) REFERENCES Airline(airline_id);
```